Title: On Cryptographic Agility Using Code-based Encryption

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Abstract:

Cryptographic agility remains a critical challenge for the NIST standardization process, particularly in satellite quantum links. This paper proposes a framework built on code-based encryption that explicitly accounts for crypto-inventory discovery in enterprises. We evaluate the framework on a simulated 100 km fiber QKD channel and compare it against 6 baseline methods. Experimental results show that the proposed approach improves secret key rate by 16.6% while keeping secret key rate within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of the NIST standardization process has changed how researchers and practitioners approach cryptographic agility, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, code-based encryption have attracted sustained attention as a promising route toward solving cryptographic agility in realistic settings such as satellite quantum links. Despite considerable progress, existing solutions for cryptographic agility tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in secret key rate frequently come at the expense of decapsulation time, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how code-based encryption interact with SLH-DSA hash-based signatures when applied to cryptographic agility. Our work is motivated by the growing demand for satellite quantum links, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of cryptographic agility, a practical design built on code-based encryption, and an empirical evaluation on a simulated 100 km fiber QKD channel. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

Subsequent work by Chen et al. (2013) extended this line of research to satellite quantum links, although the evaluation was restricted to small-scale settings. A survey by Fernandez et al. (2020) categorised existing approaches to cryptographic agility into three families and highlighted the lack of standard benchmarks. Researchers working on the NIST standardization process have explored SLH-DSA hash-based signatures as an alternative, with Takahashi et al. (2020) noting that the approach scales poorly beyond 32 components. Kowalski et al. (2016) analysed crypto-inventory discovery in enterprises and showed that purely heuristic solutions cannot guarantee consistent signature size. More recently, Lindqvist et al. (2018) combined code-based encryption with SLH-DSA hash-based signatures, achieving a 15.7% reduction in overhead on a simulated 100 km fiber QKD channel.

Several studies have criticised the reliance on synthetic workloads, arguing that results on a simulated 100 km fiber QKD channel may not generalise to satellite quantum links. In contrast to these efforts, our work treats cryptographic agility and crypto-inventory discovery in enterprises jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to cryptographic agility as a persistent obstacle for satellite quantum links. Despite this body of literature, no prior study has systematically compared code-based encryption and SLH-DSA hash-based signatures under identical experimental conditions.

3. Methodology

We formalise cryptographic agility as a constrained optimisation task in which the objective is to maximise secret key rate while keeping secret key rate within a prescribed budget. All experiments were conducted on a simulated 100 km fiber QKD channel, which provides 64 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals. Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data.

The core routine applies code-based encryption to reduce the effect of crypto-inventory discovery in enterprises, followed by a calibration step that stabilises secret key rate. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 6 baseline methods, including a conventional approach built on SLH-DSA hash-based signatures and two recent variants reported in the literature. The proposed design improves secret key rate by 16.6% relative to the strongest baseline and reduces secret key rate by 14.2%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the SLH-DSA hash-based signatures component reduces secret key rate by approximately 12.5%, confirming its contribution. A sensitivity analysis over 1200 parameter settings shows that performance remains stable, with variation in logical qubit count below 3.4%. Threats to validity include the limited diversity of a simulated 100 km fiber QKD channel and the possibility that deployment in satellite quantum links introduces conditions not captured in our setup.

4. Conclusion

This paper investigated cryptographic agility through the lens of code-based encryption, and demonstrated that a carefully designed pipeline can improve secret key rate by 16.6% on a simulated 100 km fiber QKD channel. The findings suggest that the NIST standardization process will continue to benefit from closer integration of code-based encryption and SLH-DSA hash-based signatures, particularly in satellite quantum links. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of crypto-inventory discovery in enterprises. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for cryptographic agility.